

Polycystic Ovary Syndrome

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INTRODUCTION

In 1935, Stein and Leventhal¹ described seven amenorrheic women with enlarged, polycystic ovaries who resumed cyclic menses after ovarian wedge resection. Ten years later, in a progress report Stein² emphasized for the first time the association of hirsutism and obesity with the original clinical stigmata and so defined what came to be known as Stein-Leventhal syndrome. Subsequently, the unique appearance of polycystic ovaries in women with this disorder has contributed to what today is known as polycystic ovary syndrome (PCOS).

The basis for these clinical features stems primarily from physiological alterations of ovarian function, although several other reproductive-metabolic abnormalities exist. Hirsutism results from increased theca cell androgen production. However, although increased serum adrenal androgen levels have been demonstrated in a small percentage of PCOS women. Anovulation has been attributed to decreased follicle stimulating hormone (FSH) secretion and failure of follicle maturation. The morphogenesis of the polycystic ovary remains unknown. Although approximately 50% of PCOS women are obese.³ In addition, most individuals with PCOS have insulin resistance with compensatory hyperinsulinemia.

In adults, criteria for the diagnosis include clinical or biochemical evidence of hyperandrogenism, chronic anovulation, and polycystic ovaries or any two of these three traits. However, hyperandrogenism also has been regarded as an essential feature of PCOS.

The emergence of PCOS commonly occurs at or soon after puberty. As a result, the diagnosis in young adolescent girls with irregular bleeding may be difficult. Persistence of bleeding, particularly with the gradual appearance of hair growth is highly suggestive of PCOS.

Despite the well characterized clinical presentation of PCOS, the underlying pathophysiology of PCOS remains an enigma. This chapter summarizes current understanding of PCOS and provides diagnostic and therapeutic approaches to the chronically anovulatory woman with clinical or biochemical signs of hyperandrogenism.

CLINICAL PRESENTATION

Hirsutism

Excessive hair growth is the most distinctive and visible feature of PCOS. The rate of hair growth is important as a gradual increase indicates a functional etiology, whereas rapid growth suggests an androgen producing tumor. In PCOS, increased hair growth usually occurs on the face, upper lip, and chin. In addition this hair growth may be seen on the lower abdomen as an extension of pubic hair towards the umbilicus. More severe cases include the appearance of hair on the chest. Coexisting conditions that alter the bio-activity of androgens, such as hypothyroidism and obesity, may also give rise to excessive hair growth. These conditions are associated with lowered sex hormone-binding globulin (SHBG), which provides increased availability of free testosterone.

Menstrual irregularity

In PCOS, menstrual dysfunction includes irregular, infrequent, or absent menstrual bleeding. In particular, the bleeding is unpredictable. Typically, this pattern of bleeding is an extension of postmenarchal irregularity and monthly menstrual cyclicity is never established. In some women, the onset of chronic anovulation emerges beyond adolescence, but this is unusual. Irregular bleeding is a consequence of excessive endometrial proliferation due to unopposed chronic estrogen secretion. The thickened endometrium is prone to superficial sloughing or tissue breakdown in response to persistent secretion or spontaneous decreases in circulating estrogen, respectively. The physiological link between irregular menses and anovulation relates to the persistence of estrogen production arising primarily from extraglandular conversion of androgens to estrogens. It has been assumed that within the PCO, granulosa cells generate very little estrogen because there is a lack of mature follicle development. Prolonged heavy bleeding should raise consideration of abnormal endometrial hyperplasia and even endometrial adenocarcinoma. Recognition of normal ovulation in PCOS is significant in that a history of regular menstrual cycles does not exclude the diagnosis.

Obesity

Early studies demonstrated that in PCOS obesity was present in a little more than 50% of cases.³ Currently, the rate of obesity in PCOS women has not been determined and there is a growing impression that the incidence may be greater, at least in the United States, than previously described. Commonly, an increase in the upper body or central distribution of fat gives rise to an increased waist to hip ratio. This fat distribution pattern can be found in other hyperandrogenic states, diabetes, and hyperlipidemia. As to whether PCOS patients are predisposed to obesity has not been established.

Insulin resistance

The prevalence of insulin resistance in PCOS has been reported to range between 20 and 40%.^{4, 5} That insulin resistance is common in obesity may account, in part, for the rather wide prevalence. Nevertheless, independent of obesity, the presence of a defect in insulin action in PCOS has been clearly established.⁶ Generally, the degree of insulin resistance is mild, although the prevalence of glucose intolerance and subsequent diabetes has been reported to be as high as 31% and 7.5%, respectively.^{7, 8} Notwithstanding the increased risk for diabetes, there is indirect evidence to indicate that insulin resistance may worsen the clinical manifestations of PCOS. That insulin responses following a glucose load are exaggerated in women with PCOS belies a distinct abnormality of beta cell function in this disorder. It has been reported that a subset of PCOS women with a family history of diabetes had a disposition index (measure of glucose disposal) that was in the eighth percentile, whereas in those without a similar family history, the index was in the 33rd percentile.⁹ Thus, PCOS women are prone to insulin resistance as well as altered beta cell function.

Acne

Women with PCOS may experience increased skin oiliness secondary to excessive stimulation of the pilosebaceous unit by increased androgen production. However, increased sebaceous gland activity in PCOS is not associated with acne nor is acne correlated with increased ovarian androgen production. Therefore, as an isolated symptom, acne should not be considered a sign of PCOS.

Skin Tags

Skin tags have been associated with several disorders, many of which relate to increased insulin secretion or IGF-I production. However, there are no clinical studies that show an increased prevalence of skin tags in women with PCOS. As a result, these small papillomas are not of any clinical or diagnostic significance of women suspected of having PCOS.

Acanthosis nigricans

Acanthosis nigricans has been observed in 5–50% of hyperandrogenic women and is related to the presence and severity of hyperinsulinemia.^{10, 11} It appears as symmetrical, darkened, velvety plaques that appear most commonly on the nape of the neck, in the intertriginous areas of the body such as skin folds, and on pressure bearing surfaces such as knuckles and elbows.^{10, 12} Acanthosis nigricans is not a pigmented lesion, but originates from epidermal hyperkeratosis and dermal fibroblast proliferation. While acanthosis nigricans is considered a potential marker for insulin resistance and diabetes in adults, a similar utility for this condition in children remains to be established. In PCOS reduction of hyperinsulinemia is associated with improvement in the darkened skin areas.

Infertility

Frequently, infertility is the major presenting problem in women with PCOS. Anovulation is the primary defect responsible for the failure to achieve pregnancy in this disorder. However, there is mounting evidence that women with PCOS have a higher incidence of spontaneous pregnancy loss, the mechanism of which remains unclear.^{13, 14} It has been reported in a small series of women that the prevalence of polycystic ovaries in women with recurrent miscarriage was 56%.¹⁵ By comparison, in a larger study, it was observed that polycystic ovary morphology was not predictive of pregnancy loss among women with recurrent pregnancy loss.¹⁶ Clearly, additional research is necessary to determine the prevalence of recurrent pregnancy loss in PCOS as well as the underlying mechanism.

POLYCYSTIC OVARY

Grossly, the ovaries of most women with PCOS are bilaterally enlarged and encapsulated by a thickened outer tunica albuginea that is similar to an oyster shell in appearance and color. Beneath the thickened capsule there are numerous small antral follicles contained within the ovarian cortex which gives rise to the term polycystic ovary. Typically these follicles appear on the periphery of the ovary by ultrasonography; however, 12 or more follicles per ovary are sufficient for the diagnosis. In a small percentage of ovulatory PCOS women the ovaries retain their characteristic appearance despite the appearance of a corpus luteum following ovulation. In women with hypothalamic amenorrhea who resume normal menstrual cyclicity, initial ovarian responses to increased gonadotropin secretion may be associated with increased follicle number not unlike that of a polycystic ovary.

Histologically, the distinctive feature of the polycystic ovary is the increased number of growing (preantral) and small antral follicles that is greater than that observed in normal ovaries. In the ovarian cortex most antral follicles appeared arrested at the mid-stage of development. Fully mature Graafian follicles are rare. The mechanism for the cessation of normal follicle growth is not known. The antral follicles are lined by a thin layer of granulosa cells that appears to be undergoing degeneration. However, the cells are functionally robust and exhibit increased estrogen responses to FSH stimulation compared to those of normal women, which may, in part, account for increased risk of ovarian hyperstimulation syndrome in PCOS women undergoing ovulation induction with gonadotropin therapy. Surrounding the follicles is a thickened layer of theca cells which are responsible for excess androgen production. While increased luteinizing hormone (LH) secretion drives theca cell androgen production, it appears clear that theca cells, themselves, are more responsive to LH stimulation compared to normal theca cells.¹⁷ It has been proposed that within the theca cell the steroidogenic enzyme, CYP17, is dysregulated leading to increased expression of 17-hydroxylase and greater androgen production.¹⁸

BIOCHEMICAL FEATURES

Androgens

Androstenedione and testosterone are secreted in significant amounts by the adrenal gland and by the ovary. Circulating levels of these androgens typically are elevated in women with PCOS. Using an agonist of GnRH to effect selective inhibition of ovarian steroidogenesis in women with PCOS, Chang and colleagues¹⁹ showed suppression of testosterone and androstenedione to castrate levels. These data indicate that increased production of androgens in PCOS is derived primarily from the ovary. By comparison, serum dehydroepiandrosterone sulfate (DHEA-S), an adrenal androgen, was unaltered.

It is noteworthy that serum DHEA-S levels are elevated in 20–30% of women with PCOS.^{20, 21} They remain elevated in the face of prolonged GnRH agonist administration and are suppressible with exogenous glucocorticoids, indicating that in PCOS they are almost entirely of adrenal origin. The failure of DHEA-S levels to normalize after the long-term elimination of the ovarian contribution to sex-steroid production suggests that the abnormality is not a consequence of the abnormal steroid milieu that characterizes PCOS.¹⁹ A role of elevated adrenal androgen production in PCOS women has not been established.

Estrogens

Estrogen secretion in PCOS women is characterized by chronic secretion without the cyclic pattern that accompanies an ovulatory cycle. Serum estradiol (E_2) levels may vary in PCOS but are usually in the mid-follicular phase range of 60–90 pg/mL. In contrast, serum levels of estrone (E_1) are usually greater than those of E_2 which is the reverse of the $E_1:E_2$ ratio seen in normal women. This abnormality is due to enhanced peripheral aromatization of androgens to estrogens (androstenedione to E_1 and testosterone to E_2) in extraglandular tissues in the presence of androgen excess. The clinical relevance of chronic estrogen secretion is increased risk of endometrial hyperplasia and possibly the development of endometrial carcinoma.

Progestins

In general PCOS women are anovulatory and as a result, serum progesterone levels are low and non-ovulatory. An occasional ovulation in this group of women will result in luteal concentrations of progesterone that are similar to those of normal women. Among the progestins, it has been demonstrated in women with PCOS that 17-hydroxyprogesterone values are significantly elevated. This is due to theca cell production of this hormone which is an androgen precursor. In provocative studies of androgen production, 17-hydroxyprogesterone responses to gonadotropin stimulation served to best distinguish PCOS women from normal.

Sex hormone-binding globulin

Sex hormone-binding globulin (SHBG) is a β -globulin synthesized in the liver and that binds to circulating testosterone. Only about 1% of testosterone exists in the unbound, or free, state, and thus is bioactive.²² In women with PCOS SHBG levels are abnormally low which results in an increase of the bioactive fraction of testosterone that stimulates oily skin, acne, and hirsutism. Increased serum free testosterone may also explain hirsutism in women with normal serum total testosterone values. Serum androstenedione does not bind to SHBG. Although the decrease in SHBG levels observed in PCOS has been attributed in the past to a suppressive effect of increased circulating androgen concentrations, the increased circulating insulin levels typical of PCOS seem to be primarily responsible for the observed reduction in SHBG levels.²³

Gonadotropin secretion

Serum levels of LH are elevated in most, but not all women with PCOS. Increased LH levels are more common in lean and non-obese PCOS women than in their overweight or obese counterparts. Because LH levels may vary dramatically over short periods of time it is probably ill-advised to consider the diagnosis based on the level of LH in individual patients. In PCOS women, LH secretion is distinguished by episodic bursts that occur every 60 minutes, whereas in normal women the pulse frequency varies from every 90 minutes in the early follicular phase to every 2–

4 hours in the mid-luteal phase of the menstrual cycle. The LH secretion pattern in PCOS probably reflects the unrestrained activity of the hypothalamic GnRH pulse generator which appears to be insensitive to negative feedback by progesterone.²⁴

Serum concentrations of FSH are usually normal or slightly low in PCOS. Attempts to characterize abnormal pulsatile release of FSH in PCOS women have been unrewarding because the relatively long serum half-life of FSH renders the identification of individual pulses impractical. Both gonadotropins circulate in multiple forms, which may vary in bioactivity. An elevated ratio of bioactive to immunoreactive LH has been reported in some women with PCOS, but the importance of this finding is not clear.²⁵

Prolactin

In one large series, 13% of women with PCOS were found to have prolactin levels greater than 25 ng/mL in the absence of radiographic evidence of a prolactin-secreting pituitary adenoma.²⁶ The mean prolactin level in this group was 31 ng/mL. Because prolactin secretion by the anterior pituitary is primarily under the inhibitory control of dopamine secreted into the portal circulation, it has been suggested that the hyperprolactinemia seen in some women with PCOS implies a defect in hypothalamic dopaminergic activity. Clinical studies have demonstrated that dopamine infusion lowers LH secretion;²⁷ thus, decreased dopamine activity in the hypothalamus may explain the increased LH secretion in women with PCOS. Alternatively, estrogen secretion in PCOS is chronic and unopposed by progesterone. Chronic exposure to estrogen stimulates lactotrope growth and prolactin production, which probably accounts for the occasional increased levels observed in PCOS. Hyperprolactinemia also is associated with increased adrenal production of DHEA-S, but in general the elevated adrenal androgen production seen in women with PCOS does not correlate with hyperprolactinemia.

Insulin secretion

Both obese and nonobese women with PCOS exhibit insulin resistance and release increased quantities of insulin in response to a standard glucose challenge compared with weight-matched eumenorrheic women.^{28, 29, 30} The degree of insulin resistance is subtle and not infrequently fasting insulin levels are normal. However, a normal fasting insulin level does not preclude the existence of insulin resistance. While women with PCOS are generally euglycemic there is convincing evidence that they are at risk for subsequent development of type 2 diabetes mellitus. *In vitro* testing reveals that insulin receptor number and binding are normal; however, an alteration in activation of the insulin receptor inhibits the action of insulin. Insulin signaling is conveyed through tyrosine phosphorylation of its receptor. In PCOS women, the insulin receptor

is serine phosphorylated which causes insulin resistance and compensatory hyperinsulinemia.³¹ As discussed subsequently, abnormalities of insulin secretion may have a key role in the pathophysiology of PCOS.

PATHOPHYSIOLOGY

The precise mechanism by which PCOS occurs has not been established. Current studies suggest that excess androgen production may induce polycystic ovarian morphology and perpetuate the endocrine disruption of this disorder. It has been shown that monkeys treated with subcutaneous androgen implants exhibited changes in ovarian morphology, such as increased size, thickened capsule, and increased numbers of follicles, that were not dissimilar to the ovaries observed in women with PCOS.³² These findings are relevant to the description of ovaries found in hyperandrogenic women with 21-hydroxylase deficiency and female to male transsexuals treated with chronic, high-dose androgens.^{33, 34, 35} Additionally, in the monkey model androgen administration increased the expression of FSH mRNA in granulosa cells. In PCOS women undergoing unstimulated cycle IVF, FSH binding in granulosa cells was increased compared to cells of size-matched follicles from normal women.³⁶ Taken together, these data suggest that in PCOS androgen excess may drive increased FSH receptor expression in granulosa cells which may, in part, be responsible for the heightened estrogen responsiveness to FSH stimulation during ovulation induction.

Increased androgen production may also contribute to inappropriate gonadotropin secretion in PCOS. Recent studies have revealed that high frequency LH release in PCOS women normalizes in response to physiological doses of estrogen and progesterone following treatment with an antiandrogen.²⁴ This suggests that androgen blocks the steroid negative feedback effect on gonadotropin release. Thus, hyperandrogenemia in PCOS women may cause or, at least, be one factor that causes increased LH pulse frequency in this disorder. Further studies are warranted to clearly define the mechanism of increased gonadotropin releasing hormone (GnRH) pulse generator activity in PCOS.

If hyperandrogenemia is pivotal to the pathophysiology of PCOS, then the mechanism by which androgen excess occurs is critical. Previous studies have clearly demonstrated that the theca cell is the primary source for androgen production in this disorder. In addition to increased LH secretion, the theca cell also exhibits increased responsiveness to LH stimulation.³⁷ The mechanism involves increased 17-hydroxylase activity which enhances conversion of progesterone to 17-hydroxyprogesterone. This steroidogenic enzyme is encoded by the gene, CYP17, which also encodes for 17-20 lyase, the enzyme that advances conversion of 17-hydroxyprogesterone to androstenedione. In PCOS women, 17-hydroxyprogesterone responds to GnRH agonist

stimulation were markedly higher than those of normal women, whereas androstenedione and testosterone responses were not clearly distinct.³⁸ Moreover, GnRH-stimulated 17-hydroxyprogesterone levels in PCOS were commensurate with those of normal men. The results strongly suggest that a primary defect of theca cell function also comprises the mechanism of androgen excess in PCOS. Moreover, 17-hydroxyprogesterone has become the “marker” hormone for provocative testing that assesses the capacity of the ovary to produce androgens.

As previously mentioned, the bioavailability of testosterone is dependent on SHBG. Decreases in serum SHBG are seen in women with hyperinsulinemia and hyperandrogenism. In women with PCOS, these conditions may coexist and amplify the clinical manifestations of excess androgen production. The impact of SHBG has been reported in a woman who exhibited high serum free testosterone levels and severe hirsutism during pregnancy. She was found to lack the capacity to produce bioactive SHBG due to a single nucleotide polymorphism within an allele that encoded a missense mutation.³⁹

There is clear evidence to suggest a major role of insulin resistance and hyperinsulinemia in PCOS. As to whether insulin resistance constitutes a primary mechanism has yet to be established. *In vitro*, insulin facilitates LH-induced androgen production from normal and, to a greater extent, PCOS theca cells.^{40, 41} This is consistent with improvement in insulin sensitivity and lowered serum androgens without changes in circulating LH levels following administration of insulin lowering drugs.⁴² *In vitro* and *in vivo* studies of granulosa cell function have shown that insulin enhanced estradiol responses to FSH.^{43, 44} However, recent clinical findings have suggested that PCOS granulosa cells may actually be insulin resistant as indicated by significantly increased estradiol responses to FSH during insulin infusion in PCOS women treated with a thiazolidinedione compared to responses observed prior to treatment.⁴⁵ Further investigation is necessary to resolve these conflicting results. Insulin appears not to influence pituitary LH and FSH release. Changes in gonadotropin secretion were not observed in normal or PCOS women receiving long-term insulin infusion.⁴⁶ Collectively, these data suggest that effects of insulin resistance in PCOS are directed primarily, but not exclusively, at the ovary. Hyperinsulinemia may also lower serum SHBG to increase bioactive testosterone. However, not all women with PCOS have been found to exhibit abnormal insulin secretion. As importantly, cellular defects in insulin signaling and insulin related cell function have not been consistently demonstrated within similar or different tissues of women with PCOS.

While insights into the abnormalities that comprise PCOS have advanced, several components of altered physiology remain unexplained. However, evidence is mounting that excess androgen exposure has a crucial role in the perpetuation and, perhaps, pathogenesis of this disorder.

DIFFERENTIAL DIAGNOSIS

All women with clinical features of PCOS warrant diagnostic evaluation to rule out other pathologic states that can present with similar findings. In general, these conditions include those that are associated with excess androgen production or increased testosterone bioactivity.

Ovarian hyperthecosis

Hyperthecosis is a rare diagnosis in which excess androgen is produced by nests of luteinized theca cells scattered throughout the ovarian stroma.⁴⁷ The extent of theca cell involvement may vary from minimal to extensive. In severe hyperthecosis, the ovary is slightly enlarged and extremely dense due to extensive fibroblast proliferation and is usually devoid of antral follicles. These characteristics are clearly distinct from those found in PCOS. The degree of hyperthecosis is not correlated to the severity of disease.⁴⁸ As a result, hyperthecotic tissue may be hyperresponsive to gonadotropin stimulation which would be consistent with the observation that serum LH levels are commonly normal in these women. Serum testosterone levels usually exceed those observed in women with PCOS, but are less than those found in women with androgen producing tumors. With marked increases in serum androgen concentrations, these individuals have severe hirsutism and many will exhibit virilization, e.g., clitoromegaly, temporal balding, a male body habitus, and a deepening of the voice. The androgen production may be resistant to conventional forms of long term ovarian suppression such as oral contraceptive therapy, although administration of GnRH agonists has been shown to dramatically decrease androgen production. There usually is marked insulin resistance with substantial elevations of circulating insulin levels. In addition, these patients are often obese and exhibit acanthosis nigricans.

Congenital adrenal hyperplasia

Defects in steroidogenic enzymes comprise the spectrum of congenital adrenal hyperplasia (CAH). The incomplete form of 21-hydroxylase deficiency is most common and the clinical presentation most likely mimics PCOS. The typical features include severe hirsutism, clitoromegaly, irregular menstrual cycles, and short stature. The disorder is heritable and transmitted by an autosomal recessive inheritance pattern. Morphologically, the ovaries are polycystic. A deficiency of 21-hydroxylase causes an accumulation of 17-hydroxyprogesterone and increased androgen production. An elevated basal level of 17-hydroxyprogesterone suggests the diagnosis and confirmation is made with an adrenocorticotropin hormone (ACTH) stimulation test. In selected cases, a molecular diagnostic approach to 21-hydroxylase deficiency is available which employs polymerase chain reaction to detect the genetic defect.^{49, 50} The second most common enzyme deficiency is 11 β -hydroxylase which gives rise to a mild hirsutism due to increases in 17-hydroxyprogesterone as well as 11-deoxycortisol, the immediate precursor for this enzyme. The accompanying hypertension often distinguishes this disorder from the 21-hydroxylase form of CAH.

Cushing's syndrome

Cushing's syndrome is a consequence of excessive cortisol production by either an adrenal tumor or from excessive ACTH production. In unusual circumstances the cause is iatrogenic. In most cases of ACTH overproduction the etiology is a pituitary tumor. However, rarely, ectopic sources of ACTH may be responsible as in adenocarcinoma of the lung. Clinical findings include obesity, hirsutism, acne, and menstrual irregularity. These may also be encountered in women with PCOS. However, additional evidence of moon-like facies, buffalo hump, hypertension, muscle wasting, abdominal striae, and osteoporosis indicate a primary problem of cortisol excess. While circulating androgen levels are elevated, there is also abnormal cortisol secretion characterized by increased basal levels, loss of circadian rhythmicity, and failure of suppression in response to dexamethasone. In contrast to CAH, careful examination of the ovaries does not reveal changes typical of PCOS in the vast majority of cases.

Hypothyroidism

Hypothyroidism is associated with chronic anovulation and mild hirsutism. Actually, increased hair growth may be very mild and resemble lanugo hair. The hirsutism is probably derived from depressed SHBG levels associated with hypothyroidism. The mechanism responsible for anovulation is unclear. Patients with PCOS who have normal thyroid-stimulating hormone levels should not be treated with thyroid supplementation.

Androgen producing neoplasms

Androgen producing tumors usually arise from the ovary or adrenal gland. In contrast to the gradually evolving clinical presentation associated with functional hyperandrogenism, excess androgen production by a neoplasm can be quite dramatic. Thus, careful historical documentation of the progression of symptoms, in particular, hair growth is vitally important. Within a matter of months these lesions may induce severe hirsutism, a male body habitus, and virilization marked by clitoromegaly. In addition, there may be acne and a lowering of the voice. Despite the severity of androgenic manifestations, the early stages of development of these tumors can mimic PCOS or other functional hyperandrogenic syndromes. Occasionally, the hormone production may be mixed to include excess production of cortisol and progesterone. Disruption of menstrual cyclicity varies from irregular bleeding to amenorrhea. The rapid onset of symptoms provides an important clue to the diagnosis. In some instances, a pelvic or abdominal mass can be palpated, which suggests an ovarian tumor.

EVALUATION

In general, the diagnosis of PCOS may be largely based on clinical history of hirsutism and irregular menstrual bleeding. The appearance of hirsutism is a gradual process which, unlike the rapid onset of hair growth that accompanies an androgen producing neoplasm, reflects the progressive influence of relatively moderate hyperandrogenemia. Episodes of irregular bleeding are not typically preceded by premenstrual symptomatology and, thus, are not predictable. This important clinical observation is highly suggestive of anovulation. This is not to imply that the presence of regular menses eliminates the possibility of PCOS as individuals with mild features of this syndrome have been reported to exhibit normal ovulation. Notably, the onset of these symptoms usually occurs at the time of puberty.

In most cases of PCOS the manifestation of hyperandrogenism is limited to hirsutism on the face and chin, although dark pigmented hair may occur over other regions such as the lower abdomen, chest, back, and extremities. However, extreme expression of androgen excess, such as virilization and clitoromegaly, are not typical findings in this functional syndrome. To a great degree, the variation is due to the severity and duration of hyperandrogenism as well as the ethnic background of the patient. Obesity is a distinct feature and probably occurs in the majority of PCOS patients. The obesity of PCOS is characterized by an increased waist to hip ratio. The identification of polycystic ovaries on ultrasound together with the above features is essentially diagnostic of PCOS.

The endocrine evaluation of hirsutism includes measurement of serum total testosterone (T) in the early morning hours. If the T levels are not elevated, then measurement of serum free T may be considered. Serum T and DHEA-S are not recommended as measures to exclude the possibility of an androgen producing tumor as rapid clinical progression of hair growth should distinguish a neoplastic process. If an androgen producing tumor is suspected, then corresponding threshold values for T and DHEA-S are 200 ng/dL and 7000 ng/mL, respectively.

Determination of 17-hydroxyprogesterone (17-OHP) is useful for the detection of CAH due to 21-hydroxylase deficiency. Levels greater than 2 ng/mL suggest the disorder and warrant further evaluation by an ACTH stimulation test. If Cushing's syndrome is suspected, then a 24-hour urinary free cortisol should be obtained. Values in excess of 3–4-fold above the upper normal range are suggestive of Cushing's syndrome.

Measurement of serum LH and FSH do not contribute significantly to the diagnosis of PCOS. Because of the episodic pattern of LH secretion and the apparent influence of weight on LH release, values of LH cannot be accurately interpreted. Similarly, the LH:FSH ratio adds little value in making the diagnosis.

Ultrasound imaging of polycystic ovaries is consistent with the diagnosis in women with anovulation and hyperandrogenism. In addition, demonstration of polycystic ovaries may assist in defining PCOS in women with hirsutism alone or, rarely, in those women with irregular cycles as the sole clinical feature. That the vast majority of PCOS may be determined from clinical symptomatology and polycystic ovaries may be demonstrated in normal women may make the routine use of pelvic ultrasound for the diagnosis optional. Considerable information regarding the endometrial response to chronic estrogen exposure may be achieved with ultrasound.

TREATMENT

Infertility

In general, anovulatory women with PCOS should be offered ovulation induction should pregnancy be desired. The initial consideration of therapy is clomiphene citrate with an initial dose of 50 mg/day for 5 days up to a maximum dose of 150 mg/day. This regimen may be altered depending on several factors. If obesity is significant, then a higher dose of clomiphene may be considered or, preferably, administration of an insulin lowering drug, such as metformin either alone or in combination with clomiphene. If the patient is older than 37 years, serious consideration may be given to gonadotropin therapy. Gonadotropin therapy is also beneficial for those patients unresponsive to clomiphene citrate. The increased complexity and rate of complications of gonadotropin therapy relegates its administration to individuals very familiar with this form of controlled ovarian hyperstimulation.

There has been an increasing use of aromatase inhibitors to induce ovulation in anovulatory women with PCOS. Several reports have shown that letrozole may stimulate ovulation in women with PCOS at rates comparable to that of clomiphene citrate.^{51, 52, 53, 54} The usual starting dose is 5 mg/day for 5 days with ensuing ovulation expected about a week after completion of drug ingestion. As an aromatase inhibitor, letrozole markedly lowers serum estradiol which is believed to interrupt estrogen negative feedback resulting in increased FSH secretion. However, the precise mechanism of action of letrozole has not been determined

In women with insulin resistance, ovulation may be achieved with the use of insulin lowering drugs, such as the biguanide, metformin.^{55, 56, 57} Reported rates of ovulation vary substantially and may reflect the diverse nature of the study populations. A recent study conducted in the United States found that metformin was relatively ineffective compared to clomiphene citrate in significantly obese women with PCOS.⁵⁸ By comparison, higher ovulatory rates have been observed in studies performed in other parts of the world.^{56, 57} To achieve ovulation, the target metformin dose is 1500–2000 mg/day. Ovulation may also be achieved with the use of thiazolidinedione derivatives, but this class of compounds are considered category C by the FDA and not recommended for infertility.

Bilateral ovarian wedge resection or electrosurgical destruction of ovarian tissue should be undertaken only as a last resort. It is not clear that a prolonged period of ovarian cyclicity can be expected after such procedures, and postoperative adhesions may interfere with subsequent conception. The reader is referred to other chapters of this library for a detailed discussion of ovulation induction techniques.

Hirsutism

The patient complaining primarily of hirsutism usually benefits from cyclic oral contraceptive administration. The combination of decreased ovarian androgen secretion and elevated SHBG levels that results from use of an oral contraceptive frequently results in reduced peripheral androgen activity and some cosmetic improvement. In severe cases or those with minimal response to oral contraceptives, an antiandrogen may be added to the treatment regimen. Spironolactone, an aldosterone antagonist, is recommended at a dose of 100–200 mg/day. It acts by inhibiting androgen production in the ovary and by competitively blocking local androgen effect on the hair follicle. Patients treated with spironolactone should be aware that they may become ovulatory. Because the effects of this antiandrogen on any developing fetus are unknown, women taking spironolactone should use contraception if sexually active.

Other antiandrogens include flutamide, a nonsteroidal antiandrogen that is potentially hepatotoxic, and finasteride, a 5- α reductase inhibitor that blocks conversion of testosterone to dihydrotestosterone. These drugs are not commonly used in the United States because of low efficacy. Cyproterone acetate is an effective antiandrogen used widely in Europe for the treatment of hirsutism. It is generally given with estrogen in a cyclic fashion. It is currently unavailable for use in the United States, but has widespread use in Europe. The antiandrogen, drospirenone, also has progestin activity and has been formulated with estrogen as an oral contraceptive. However, drospirenone is a very weak antiandrogen and the dose used in oral contraceptives, 3 mg, is roughly equivalent to 25 mg of spironolactone and 1 mg of cyproterone acetate.⁵⁹

Direct methods of hair removal include laser therapy, electrolysis, shaving, plucking, or topical creams. In general, laser therapy affords more effective treatment than electrolysis, but is short-lived and expensive. Application of a topical cream such as eflornithine cream may result in a more rapid initial response.

Menstrual dysfunction

Women with PCOS may be amenorrheic or oligomenorrheic or may have frequent, sometimes heavy, irregular bleeding, referred to as dysfunctional uterine bleeding. The fact that these patients may claim to bleed on a regular monthly basis should not be interpreted as ovulatory cycles. In all cases, except for the occasional patient who is ovulatory, the practitioner must be

mindful of chronic estrogen secretion in PCOS women and the increased risk of endometrial hyperplasia and adenocarcinoma. This risk largely can be eliminated by administration of oral contraceptives or periodic progestins. For affected women not using oral contraceptives, medroxyprogesterone acetate, 5 mg/day, or micronized progesterone, 200 mg/day, should be taken orally for 10–14 days on a monthly basis. Patients on this regimen should be advised to use a barrier form of contraception if they do not wish to conceive. Finally, the practitioner should have a low threshold for carrying out endometrial sampling in women with abnormal bleeding, particularly if they have not been provided with such therapy for an extended period of time.

Risk of type 2 diabetes mellitus

A growing body of evidence suggests that women with PCOS, particularly those who are obese, are at increased risk for the development of type 2 diabetes. Generally, type 2 diabetes is preceded by insulin resistance followed by impaired glucose tolerance. Thus, recognition of insulin resistance in PCOS women, particularly in those with obesity and a positive family history, should alert the physician to subsequent risk of diabetes. It is reasonable to screen PCOS patients for glucose intolerance periodically by obtaining fasting glucose and insulin levels. Normal results are reassuring, but do not preclude the existence of insulin resistance. In general, obese women with PCOS should be considered insulin resistant until proven otherwise. Abnormally increased fasting insulin values indicate insulin resistance. Fasting glucose levels above 100 mg/dL and less than 125 mg/dL indicate impaired glucose tolerance, whereas levels above 125 mg/dL are consistent with type 2 diabetes. In women with obesity, it is also recommended that a lipid profile and liver function tests be obtained. In these patients abnormal aspartate aminotransferase (AST) and alanine aminotransferase (ALT) levels may reflect fatty infiltration and possibly non-alcoholic steatosis.

Prevention strategies have been suggested by the Diabetes Prevention Program study which showed that individuals at risk for diabetes significantly lower the likelihood of disease by modest weight loss through diet and exercise.⁶⁰ The study also revealed that metformin administration also was beneficial, but less so than diet and exercise.

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